# Trigonometry

**CHAPTER 6**

**Height and Distance** **6.1–6.2**

DPP 6.1 Problems Based on Height and Distance 6.1

Ananth Garg

# CHAPTER 6 Height and Distance

## DPP 6.1
### Problems Based on Height and Distance

**Single Correct Answer Type**

**1.** A vertical tower $AB$ has its end $A$ on the level ground. Let $C$ be the midpoint of $AB$ and $P$ be a point on the ground such that $AP = 2AB$. If $\angle BPC = \beta$, then $\tan\beta$ is equal to

(a) 1/9 (b) 2/9
(c) 3/7 (d) 2/7

**2.** Two vertical poles of heights 20 m and 80 m stand apart on a horizontal plane. Then the height (in meters) of the point of intersection of the lines joining the top of each pole to the foot of the other from the horizontal plane is

(a) 10 (b) 12
(c) 15 (d) 16

**3.** $AB$ is a vertical pole with $B$ at the ground level and $A$ at the top. A man finds that the angle of elevation of point $A$ from a certain point $C$ on the ground is 60°. He moves away from the pole along the line $BC$ to a point $D$ such that $CD = 8$ m. From $D$, the angle of elevation of point $A$ is 45°. Then the height of the pole is

(a) $4\sqrt{3}(\sqrt{3}-1)$ m (b) $y = 4\sqrt{3}(\sqrt{3}+1)$ m
(c) $8\sqrt{3}(\sqrt{3}-1)$ m (d) $8\sqrt{3}(2-\sqrt{3})$ m

**4.** Two poles standing on a horizontal ground are of heights 6 m and 12 m, respectively. The line joining their tops make an angle of 15° with the ground. Then the distance (in m) between the feet of poles is

(a) $3(2+\sqrt{3})$ (b) $6(\sqrt{3}+1)$
(c) $6(2+\sqrt{3})$ (d) $9(\sqrt{3}-1)$

**5.** Consider a triangular plot $ABC$ with sides $AB = 6$ m, $BC = 7$ m and $CA = 8$ m. A vertical lamppost at the mid-point $D$ of $AC$ subtends an angle 30° at $B$. The height (in m) of the lamppost is

(a) $\sqrt{\frac{159}{2}}$ (b) $\sqrt{159}$
(c) $2\sqrt{159}$ (d) $\sqrt{\frac{159}{2}}$

**6.** The angle of elevation of the top of two vertical towers as seen from the middle point of the line joining the foot of the towers are 60° and 30° respectively. The ratio of the height of the towers

(a) 3 : 1 (b) 2 : 1
(c) 3 : 2 (d) 4 : 3

**7.** An 8 feet tall man finds that the angle of elevation of the top of a 26 feet high pillar and the angle of depression of its base are complementary angles. The distance of the man from the pillar is

(a) $8\sqrt{3}$ feet (b) $6\sqrt{3}$ feet
(c) 12 feet (d) $4\sqrt{3}$ feet

**8.** The angle of elevation of the top of an incomplete vertical pillar at a point 40 m away from its base on horizontal plane is 45°. If the angle of elevation of the top of the complete pillar at the same point is to be 60°, then the height of the incomplete pillar is to be increased by

(a) $40(\sqrt{3}-1)$ m (b) $40(\sqrt{3}+1)$ m
(c) $40(2-\sqrt{3})$ m (d) $20(2+\sqrt{3})$ m

**9.** Two persons on two different buildings at the same height of 10 m observe an object on the ground. If the persons and the object lie in the same vertical plane and the angle of depressions are 30° and 60°, then the horizontal distance between the persons is

(a) $\frac{10}{\sqrt{3}}$ m (b) $\frac{20}{\sqrt{3}}$ m

(c) $\frac{40}{3}$ m (d) $10\sqrt{3}$ m

**10.** A tower of 10 m height is standing vertically at some point on the arc of a semi-circular park whose diameter is $AB$. If angles of elevations of the top of the tower from $A$ and $B$ are 30° and 60°, respectively, then the radius of the arc is

(a) $5\sqrt{\frac{10}{3}}$ m (b) $3\sqrt{10}$ m

(c) $10\sqrt{\frac{10}{3}}$ m (d) $\frac{10\sqrt{10}}{3}$ m

## Answer Key

### Single Correct Answer Type

**1.** (b) **2.** (d) **3.** (b) **4.** (c) **5.** (a) **6.** (a) **7.** (c) **8.** (a) **9.** (b) **10.** (a)

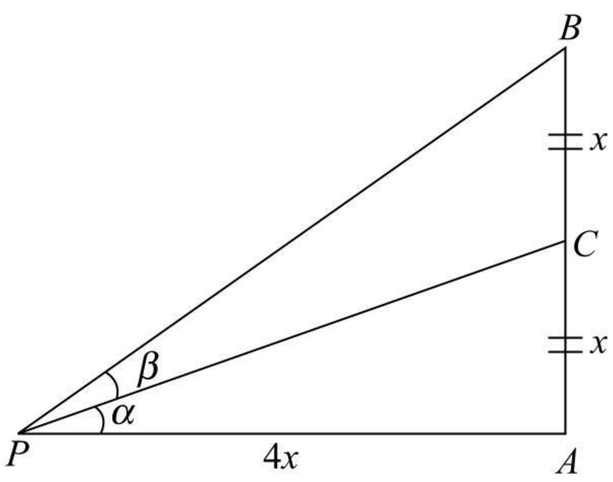
Ananth Garg

# CHAPTER 6

## DPP 6.1

### Single Correct Answer Type

**1. (b)** Consider the following diagram:

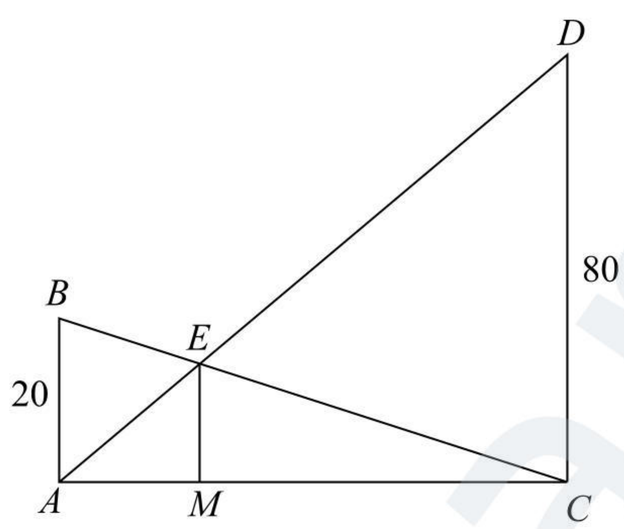



From the figure,

$$\tan(\alpha+\beta) = \frac{AB}{AP} = \frac{1}{2} \quad \text{and} \quad \tan\alpha = \frac{AC}{AP} = \frac{1}{4}$$

$$\Rightarrow \quad \frac{\frac{1}{4}+\tan\beta}{1-\frac{1}{4}\tan\beta} = \frac{1}{2} \quad \Rightarrow \quad \tan\beta = \frac{2}{9}$$

**2. (d)** Let $AB$ and $CD$ be the given towers.




Here, $\Delta ACD$ and $\Delta AME$ and similar.

$$\therefore \quad \frac{AM}{AC} = \frac{EM}{80} \qquad \ldots(1)$$

Also, $\Delta BAC$ and $\Delta EMC$ are similar.

$$\therefore \quad \frac{MC}{AC} = \frac{EM}{20} \qquad \ldots(2)$$

Adding (1) and (2), we get

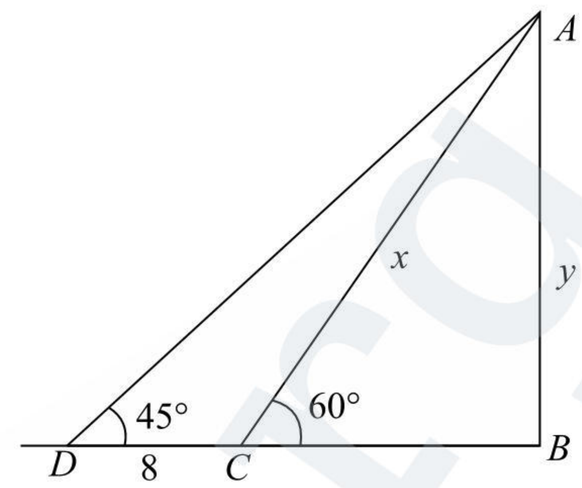
$$\frac{AM+MC}{AC} = EM\left(\frac{1}{20}+\frac{1}{80}\right)$$

$\therefore \quad EM = 16$ meters

**3. (b)** Consider the following diagram:

From the figure,

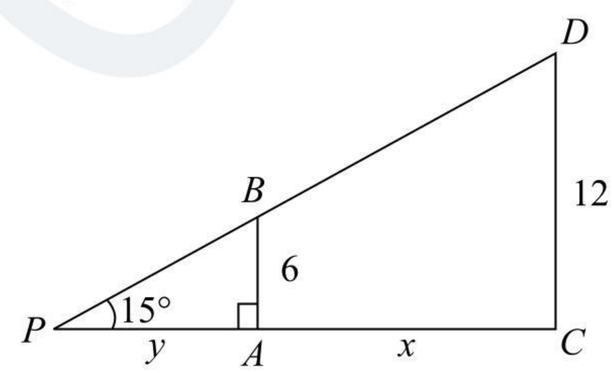
$$\tan 60° = \frac{y}{x} \quad \text{and} \quad \tan 45° = \frac{y}{x+8}$$




$$\therefore \quad 1 = \frac{y}{\frac{y}{\sqrt{3}}+8} \quad \Rightarrow \quad y = \frac{8\sqrt{3}}{\sqrt{3}-1}$$

$$\Rightarrow \quad y = 4\sqrt{3}\,(\sqrt{3}+1)$$

**4. (c)** Let $AB$ and $CD$ be the poles of heights 6 m and 12 m, respectively.



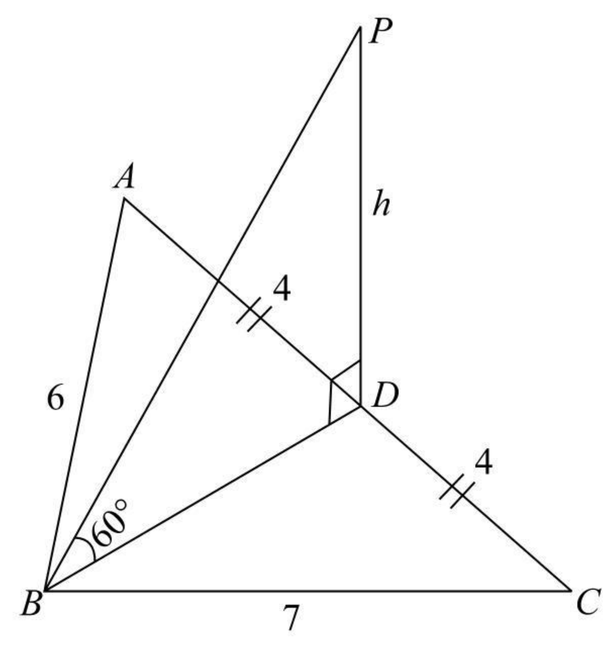

From the figure,

$$\tan 15° = \frac{12}{y+x} \quad \text{and} \quad \tan 15° = \frac{6}{y}$$

$$\therefore \quad \tan 15° = \frac{12}{\frac{6}{\tan 15°}+x} \quad \Rightarrow \quad 6 + (\tan 15°)\,x = 12$$

$$\Rightarrow \quad x = \frac{6}{\tan 15°} = \frac{6}{2-\sqrt{3}} = 6(2+\sqrt{3})$$

**5. (a)** Consider the following diagram:




In $\Delta ABC$, $BD$ is median.

So, using Apollonius theorem, we get

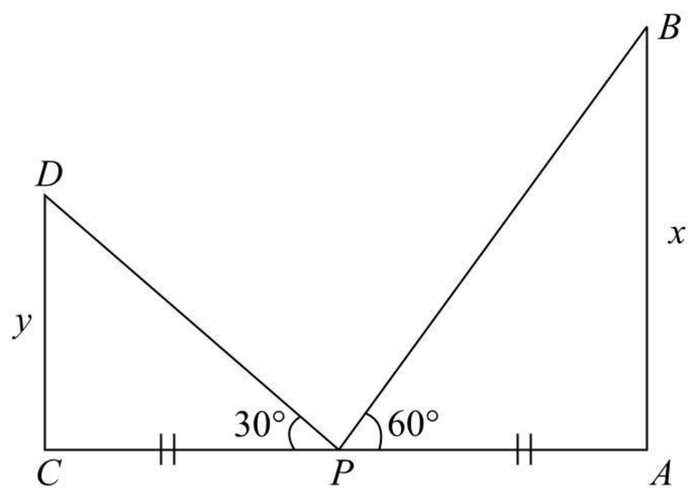
$$AB^2 + BC^2 = 2(BD^2 + CD^2)$$

$$\Rightarrow \quad 36 + 49 = 2(BD^2 + 16)$$

$$\Rightarrow \quad BD = \sqrt{\frac{53}{2}}$$

In $\Delta BDP$, $\tan 60° = \dfrac{h}{\sqrt{\frac{53}{2}}} \quad \therefore \quad h = \sqrt{\dfrac{159}{2}}$

**6. (a)** Consider the following diagram:



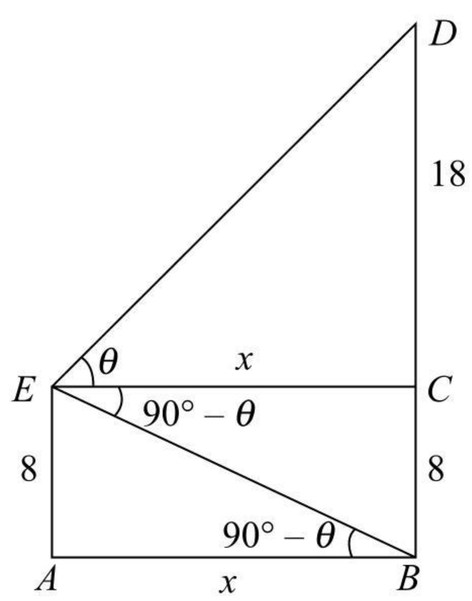

From the figure,

$$\tan 60° = \frac{x}{AP} \quad \text{and} \quad \tan 30° = \frac{y}{CP}$$

Now, $AP = CP$

$$\Rightarrow \quad \frac{x}{\tan 60°} = \frac{y}{\tan 30°} \quad \Rightarrow \quad x = 3y$$

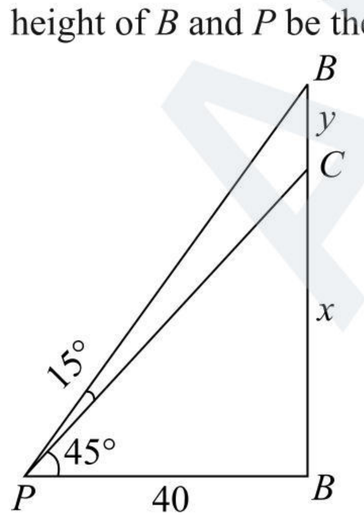
**7. (c)** Let $BD$ be the tower and $AE$ be the man.




From the figure,

$$\tan\theta = \frac{18}{x} = \frac{x}{8} \quad \Rightarrow \quad x^2 = 144 \quad \Rightarrow \quad x = 12$$

**8. (a)** Let $AC$ be the incomplete tower which needs to be taken to the height of $B$ and $P$ be the point on the horizontal plane.

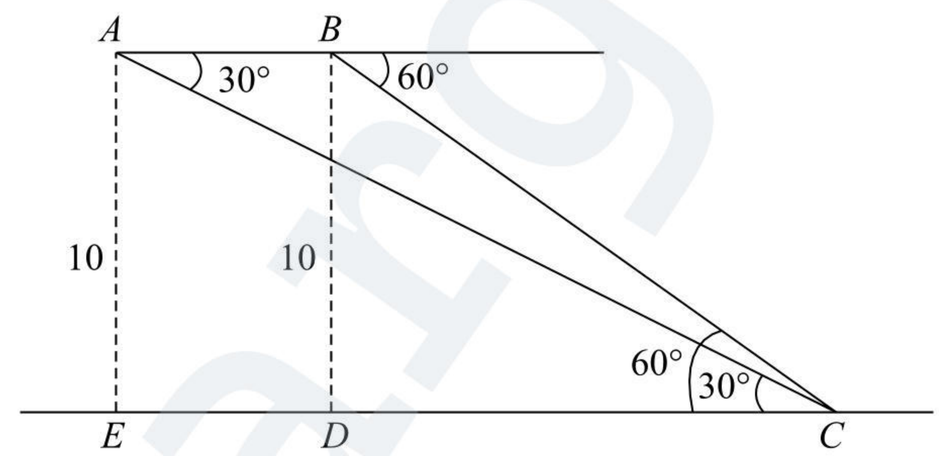



From the figure,

$$\tan 45° = \frac{x}{40} \text{ or } x = 40 \quad \text{and} \quad \tan 60° = \frac{x + y}{40}$$

$\therefore \quad y = 40\,(\sqrt{3} - 1)$ meters

**9. (b)** Let $A$, $B$ be the persons, $E$, $D$ be the feet of the towers and $C$ be the object.



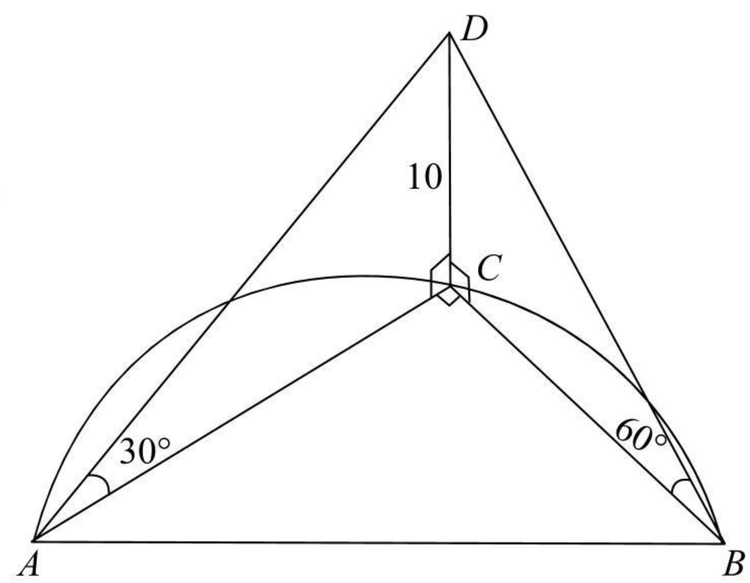

From the figure,

$CD = 10 \cot 60°$ and $CE = 10 \cot 30°$

$\therefore \quad DE = CE - CD = 10\,(\cot 30° - \cot 60°)$

$$= 10\left(\sqrt{3} - \frac{1}{\sqrt{3}}\right) = \frac{20}{\sqrt{3}} \text{ m}$$

**10. (a)** Let CD be the tower.




From the figure,

$AC = 10 \cot 30° = 10\sqrt{3}$

and $BC = 10 \cot 60° = \dfrac{10}{\sqrt{3}}$

In $\Delta ABC$, $(10\sqrt{3})^2 + \left(\dfrac{10}{\sqrt{3}}\right)^2 = AB^2$

$$\Rightarrow \quad AB^2 = 300 + \frac{100}{3} = \frac{1000}{3}$$

$$\Rightarrow \quad AB = 10\sqrt{\frac{10}{3}}$$

$$\therefore \quad \text{Radius} = 5\sqrt{\frac{10}{3}}$$